

Basic Principles of Medicine 1
Module : Foundations to Medicine
DLA

Early Embryology Week 3

Recommended Reading

Moore KL, Persaud TVN, Torchia MG. The Developing Human – Clinically Oriented Embryology, 11th Ed. Chapters 1-4. pgs. 37-39, 47 - 56

- Pay special attention to the Green boxes for the clinical correlations for this topic.

Objectives

SOM.MKII.BPM1.1.FTM.1.ANAT.1024	Describe the development of the notochord and its role in induction of the neural plate.
SOM.MKII.BPM1.1.FTM.1.ANAT.1025	Describe the fate of the intraembryonic mesoderm, including the development of the somites and intraembryonic coelom.
SOM.MKII.BPM1.1.FTM.1.ANAT.1026	Describe the folding of the embryo, paying particular attention to the sequence and timing at which folding occurs.
SOM.MKII.BPM1.1.FTM.1.ANAT.1027	Describe the embryology and clinical presentation of sacrococcygeal teratoma
SOM.MKII.BPM1.1.FTM.1.ANAT.1028	Describe the process of gastrulation and its significance in embryonic development.

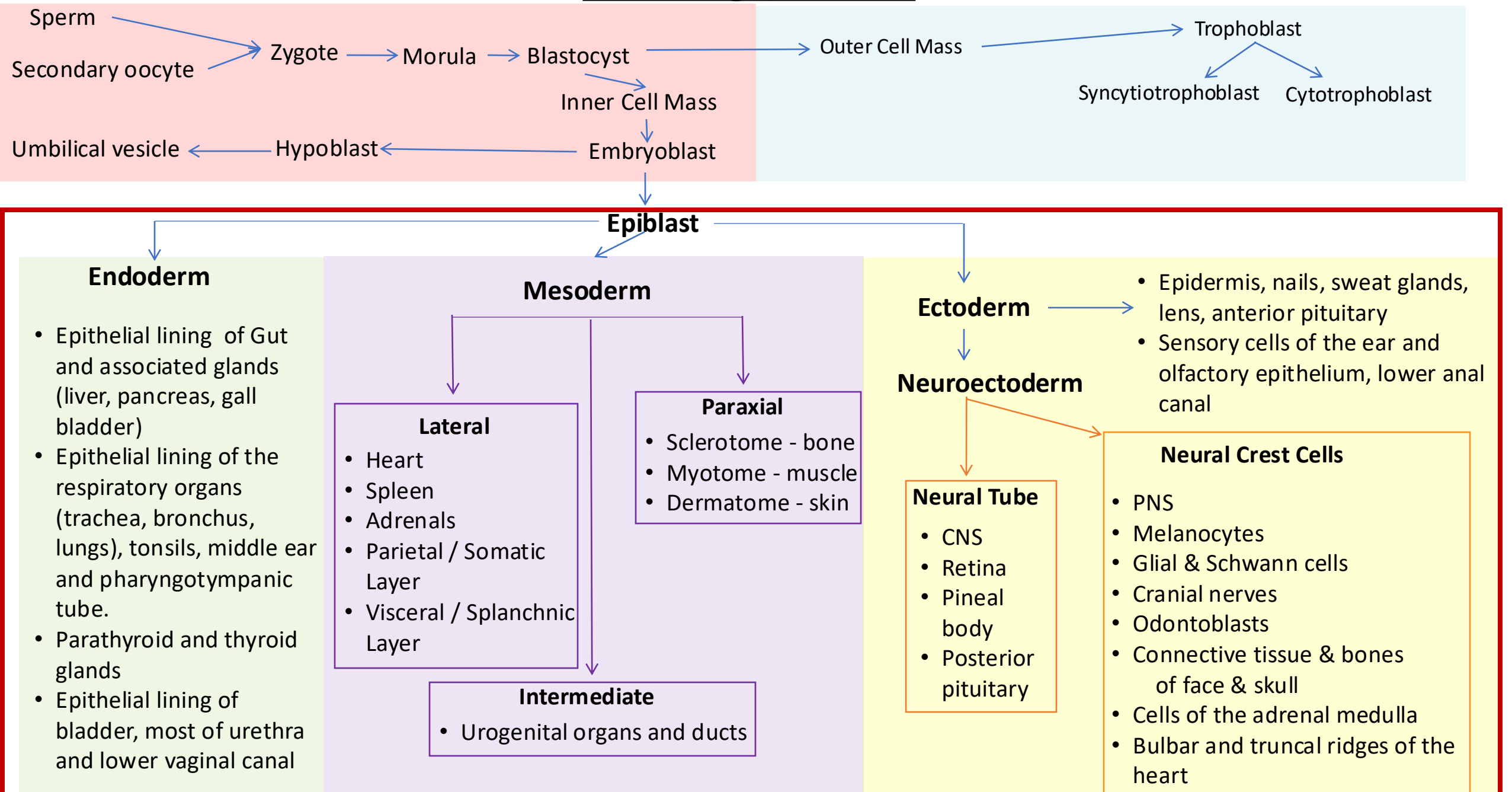
Special note

Some slides have a green background, this will not be covered in the video recording but are put so you have the information available

Terminology

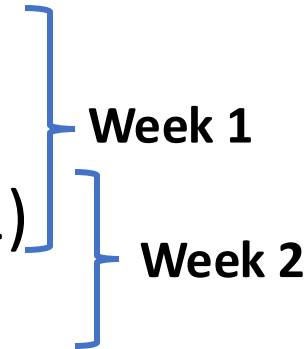
- **Proliferation:** Increase in number of cells by mitotic division
- **Differentiation:** Changing of cells from immature stem cells into functionally mature cells
- **Cell migration:** Movement of cells from one area or part of the developing embryo to another
- **Gastrulation:** The formation of the germ layers from the epiblast
- **Invagination:** Form of cell migration where cells move without detaching from each other forming a dent in the surface
- **Involution:** The inward movement of an expanding outer layer so that it spreads over the internal surface of the remaining external cells
- **Neurulation:** Formation of the neural tube
- **Germ layers/Embryonic cell lines:** Three distinct types of cells that differ in structure and function – each line gives rise to specific functional tissue types
- **Caudal:** area close to the tail
- **Cranial:** area close to the head
- **Mesenchyme:** an embryonic tissue that gives rise to supporting tissues (connective tissue) of the embryo
- **Canulation:** the process of acquiring a lumen

The Big picture



The journey

- The development of the embryo from male and female gametes to a fully formed embryo is termed Embryogenesis
- It consists of a series of events
 - Fertilization – Day 1
 - Cleavage – Day 1 - 6
 - Implantation – Day 6 – 10 (beginning of week 2)
 - Formation of the bilaminar disc – Day 10 - 14
 - Gastrulation (formation of embryonic cell lines) - Day 14 – 21 (**Week 3**)
- Organogenesis – **Week 4 to 8**
- Everything happens synchronously with many simultaneous events

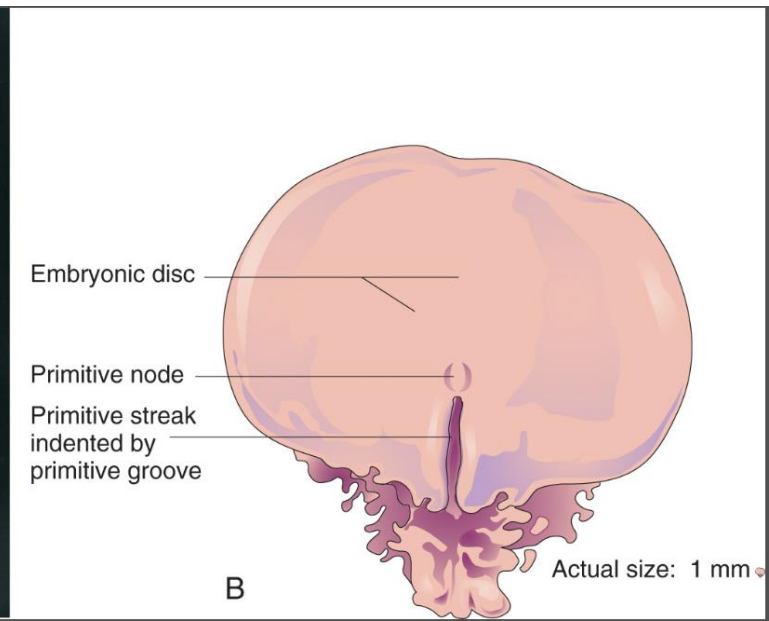
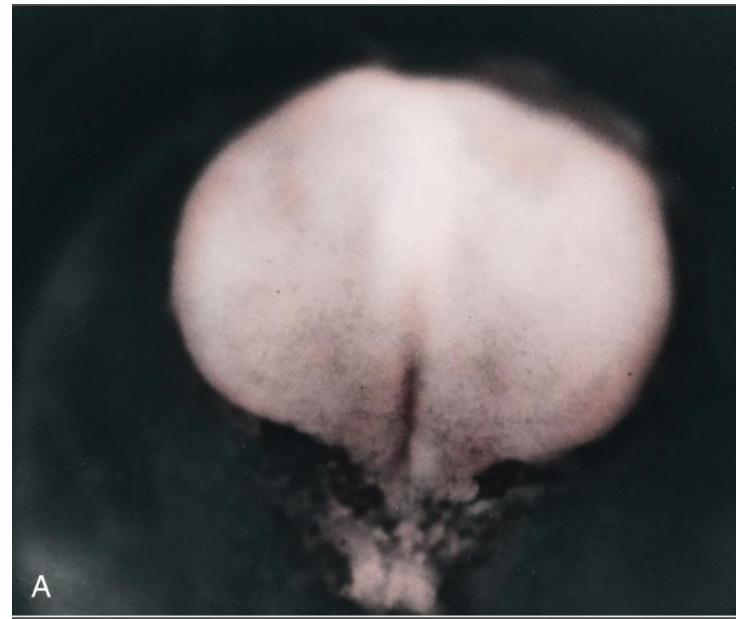
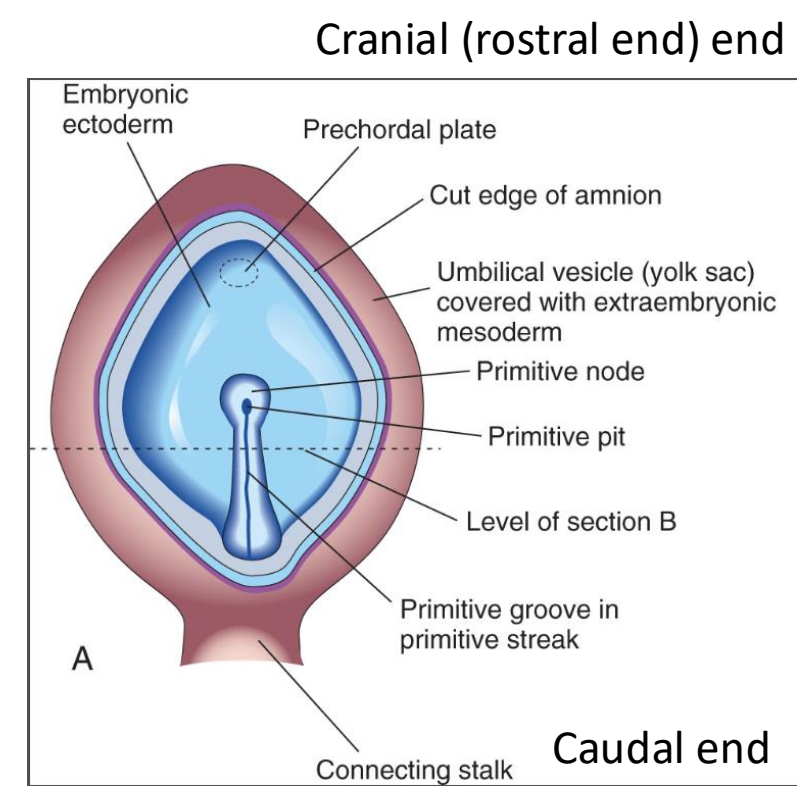


Additional things that happen in week 3

- Before and during the gastrulation process
- Several other events take place in the embryo
 - 1st is the development of blood islands in the extraembryonic mesoderm at the end of week 2
- Cells migrate intelligently toward their future adult locations
- This sets up the conditions for the start of organ formation (**organogenesis**)
- These simultaneous events are the result of gastrulation
 - Primitive blood vessels form
 - Notochord forms
 - Neural tube forms
 - Somites form
 - Blood vessels form

Week 3 - Gastrulation

- Formation of the three germ layers - **Gastrulation**
- Epiblast cells proliferate and migrate to the caudal region of the embryo
 - creates a thickened linear band = **Primitive streak**
 - Left/right, caudal/cranial and dorsal/ventral parts can now be identified
- The streak continues to form toward the caudal end by addition of cells
- Cranial end proliferates to form the **primitive node**

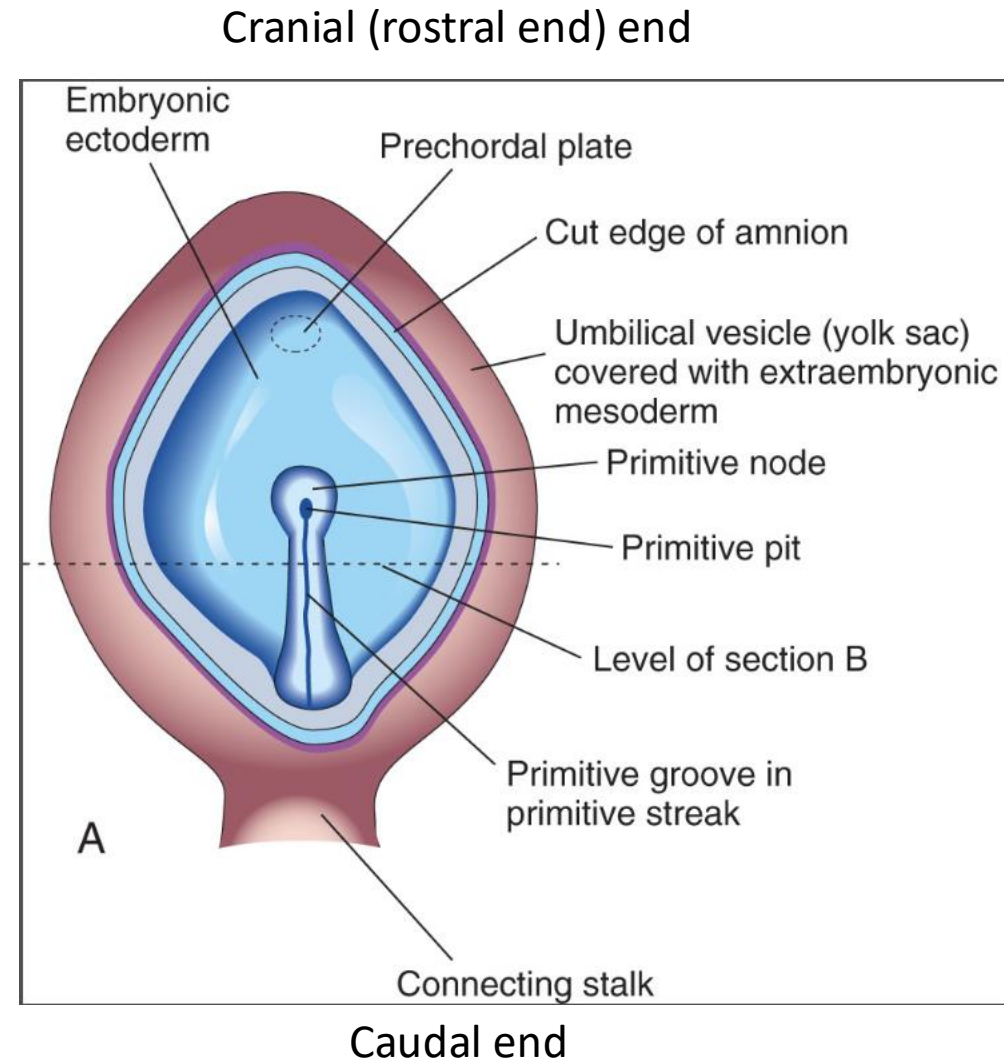


Week 3 - Gastrulation

- Invagination of epiblast cells form a narrow groove in the primitive streak = **primitive groove**
 - continuous with a small depression in the primitive node, the **primitive pit**
- Cells from the deep surface of the streak leave as **mesenchyme**

Watch a video on invagination (slow down speed to 0.25):

https://youtu.be/zOqeDABZilA?si=nrMEzc_5Vj0nvCh



Week 3 - Gastrulation cont.

- **Formation of endoderm:**

- epiblast cells migrate through the primitive streak and node displacing the hypoblast

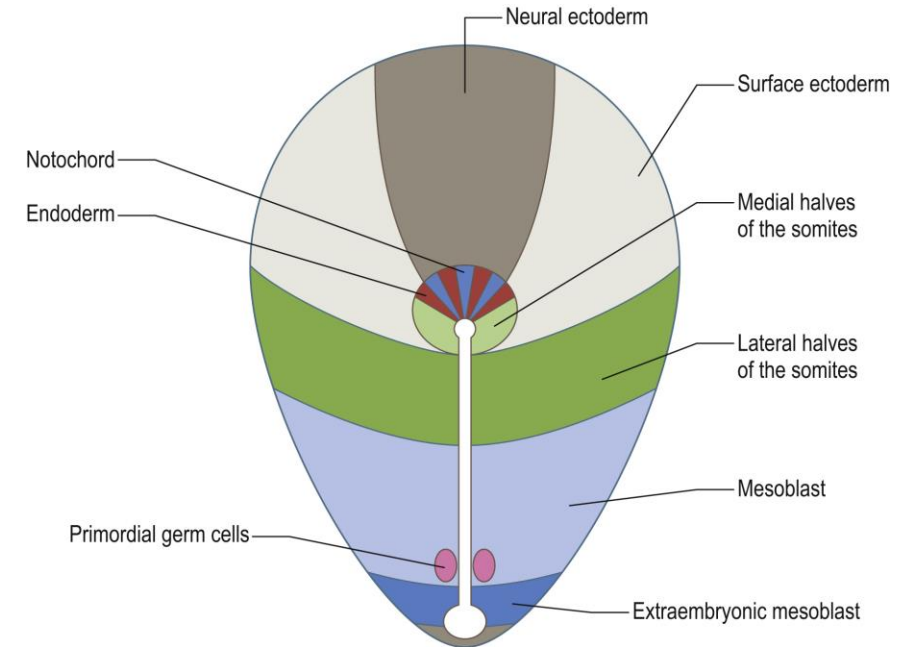
- **Formation of intraembryonic mesoderm:**

- epiblast cells migrate through the streak and lie between the epiblast and endoderm layers

- **Ectoderm:**

- epiblast cells that don't migrate

Cell fates are determined by where the cells **involute through the primitive streak**



Two locations where ectoderm and endoderm remain firmly adhered to each other (no intervening mesoderm)

- Cranially, at the prechordal plate = **oropharyngeal membrane**
- **Cloacal membrane**, caudal to the origin of the primitive streak

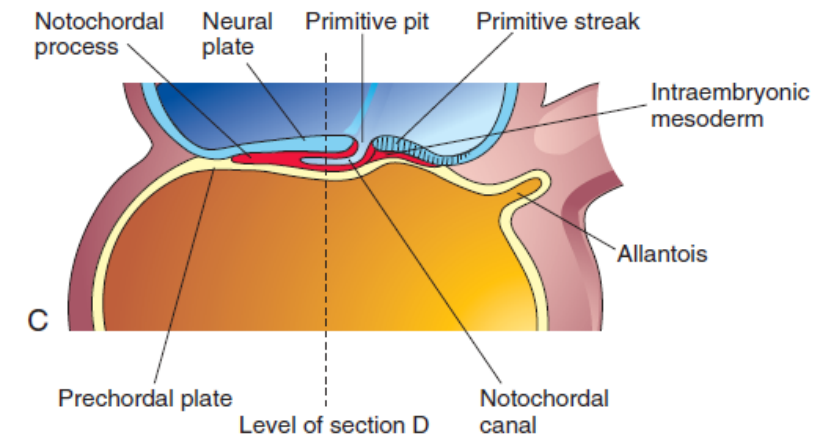
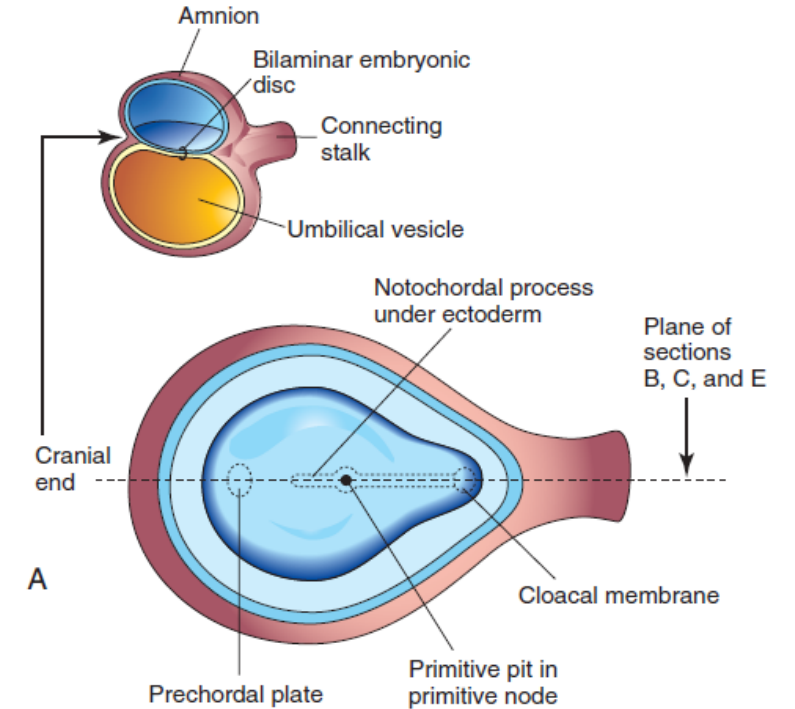
Week 3 – Formation of the notochord

- **Notochordal process:**

- Mesenchymal cells migrate cranially from and through the primitive pit
- Grows toward the **prechordal plate** (between ectoderm and endoderm)
- Acquires a lumen = **notochordal canal**

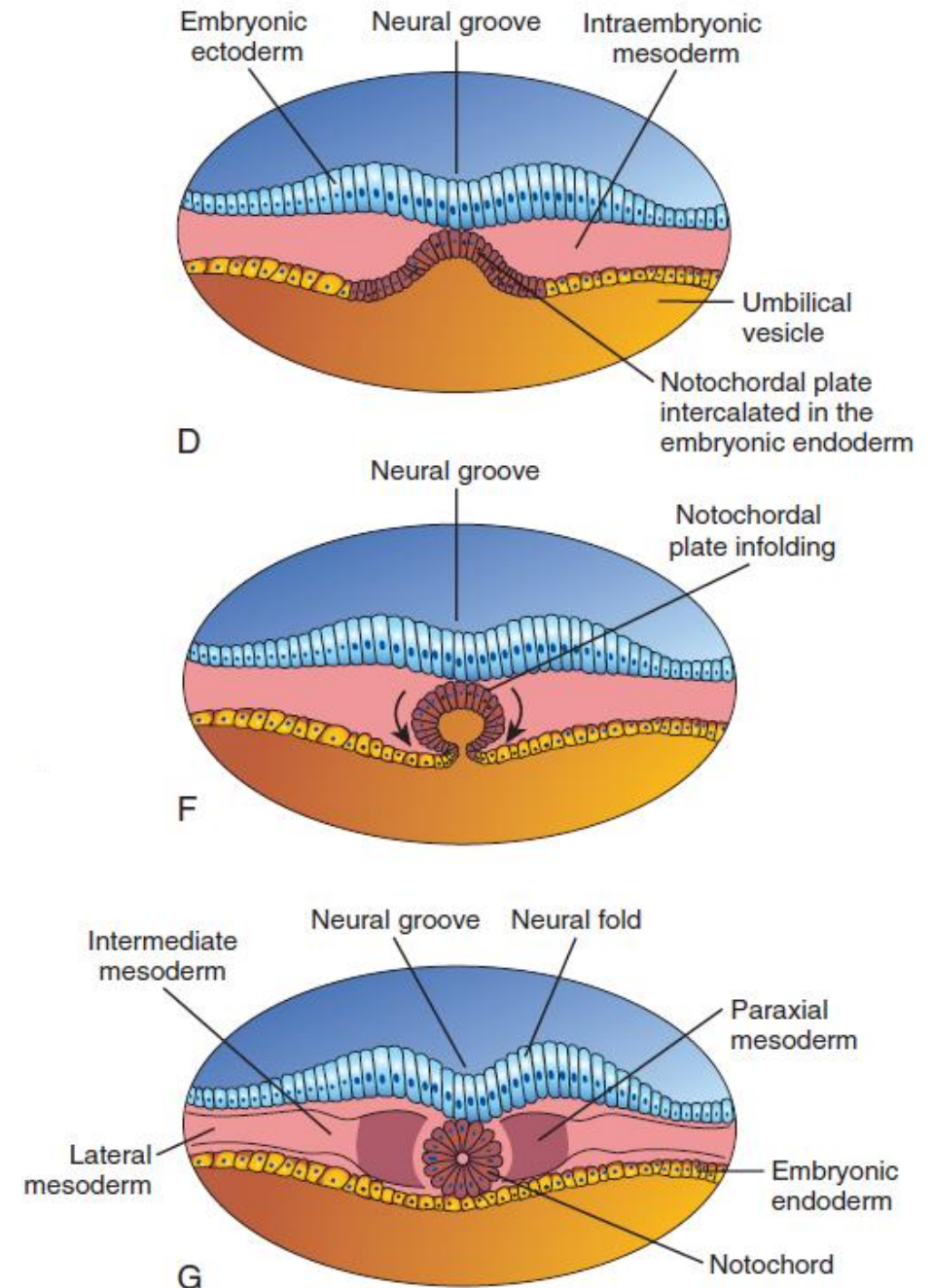
- **Notochordal plate:**

- Fusion of the floor of the notochordal process with the underlying endoderm
- Involuting toward the ectoderm
- Detaches from the endoderm and its free edges fuse = **notochord**.



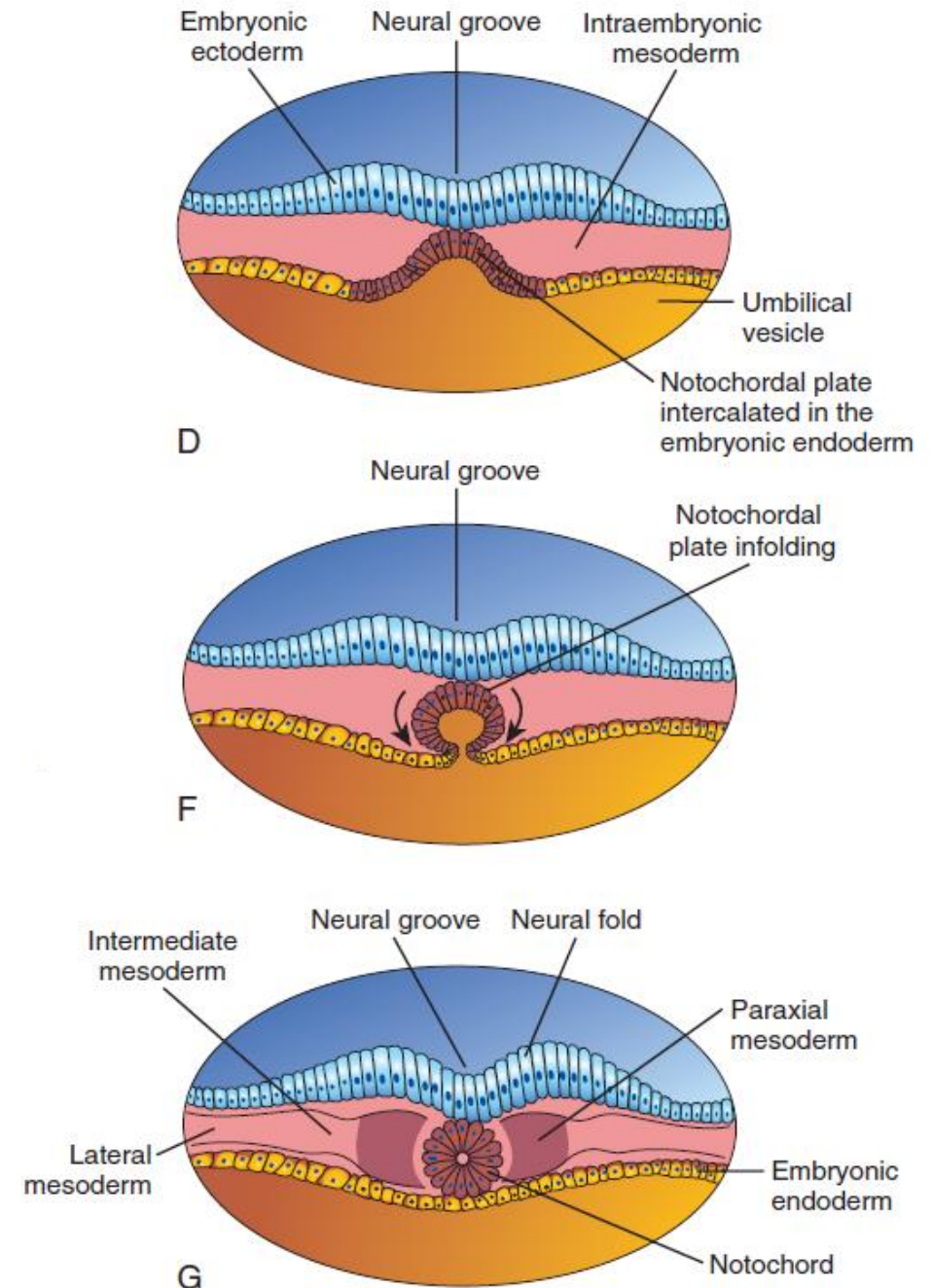
Week 3 – Notochord

- A solid cord of cells
- **Serves as the longitudinal axis of the embryo**
- Induces the overlying ectoderm cells to become elongated = **Neural plate (neuroectoderm)**
- Induces differentiation of the mesoderm into distinct cell populations
- **Contributes to the intervertebral discs (which remains in the adult)**



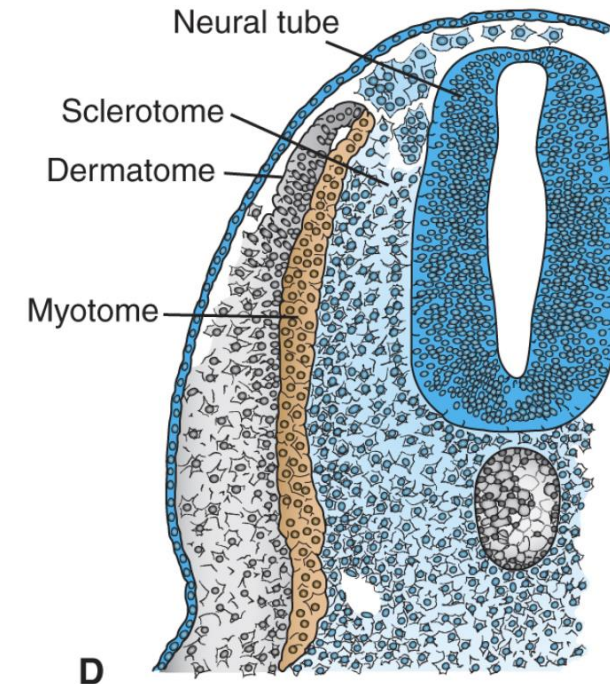
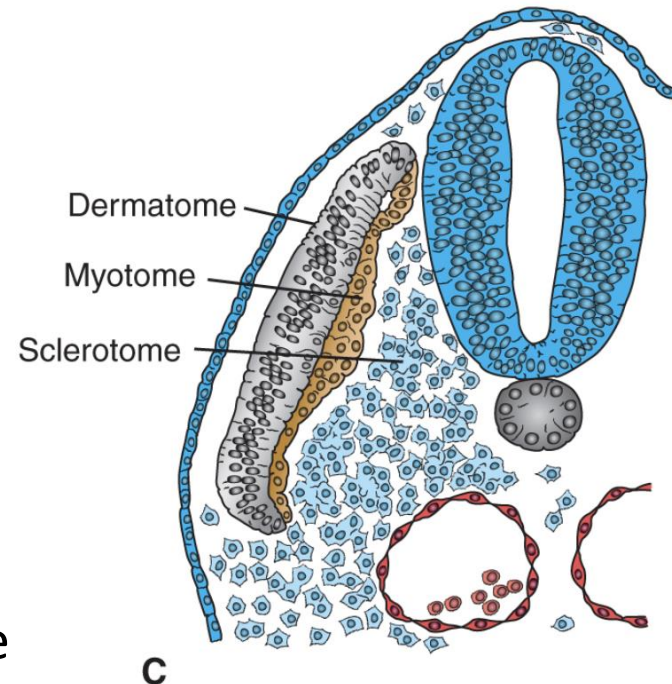
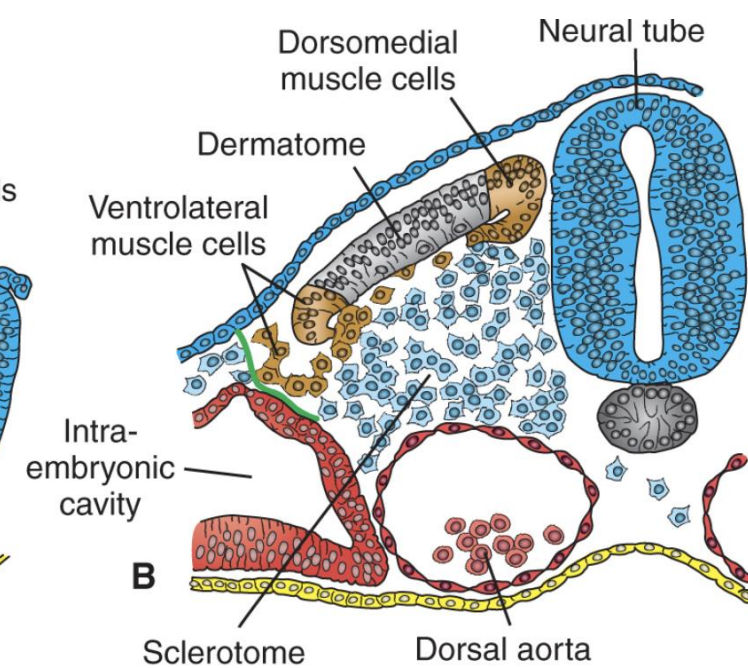
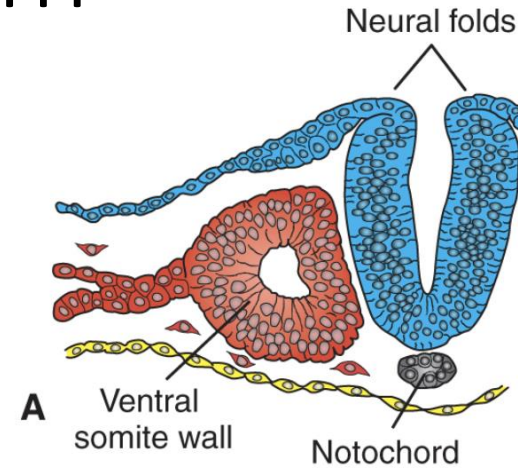
Week 3 continued...

- The **intraembryonic mesoderm** on either side of the notochord differentiates into three parts:
 - **Paraxial mesoderm**
 - **Intermediate mesoderm**
 - **Lateral mesoderm**
- Mesenchymal cells from the primitive streak and notochordal process continue to migrate
 - Some form cardiogenic tissue
 - This will eventually form the heart



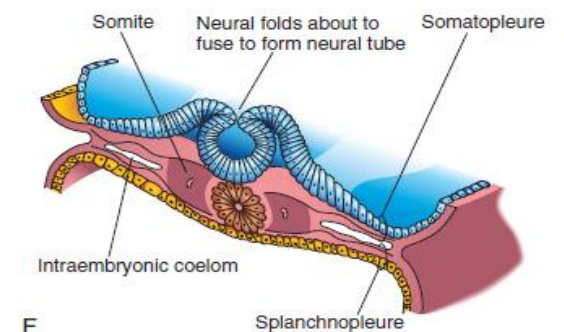
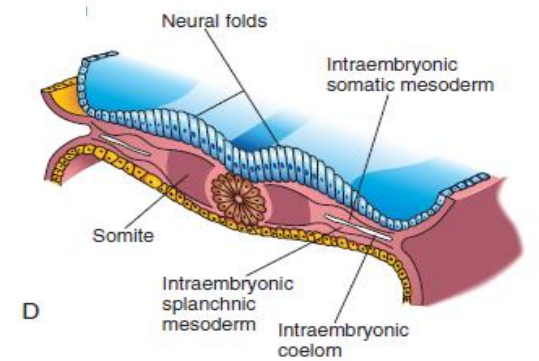
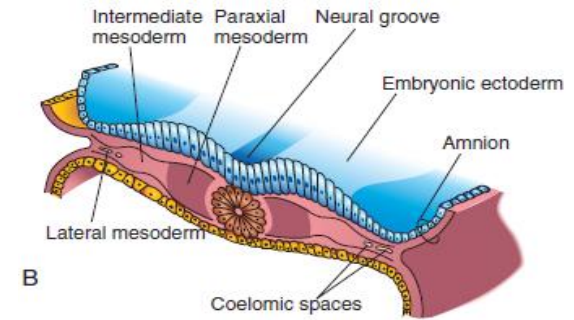
Week 3 – Paraxial mesoderm

- The paraxial mesoderm form distinct bodies = **somites**
- Identified as bumps on the surface of the developing embryo
- Each somite has three distinct portions
 - **Sclerotome** - bones
 - **Myotome** – skeletal muscle
 - **Dermatome** – dermis of skin
- Each somite is associated with a spinal nerve



Week 3 - Intermediate and Lateral plate mesoderm

- **Intermediate mesoderm:** gonads and urinary system
- **Lateral mesoderm and cardiogenic mesoderm**
 - Clefts appear in the lateral and cardiogenic mesoderm which coalesce to form **intraembryonic coelom (IC)** splitting the LPM into two layers
 - **Parietal layer or somatopleure** - formation of the body wall
 - **Visceral layer or splanchnopleure** – formation of organs
 - Intraembryonic coelom further divides into three body cavities:
 - **Pericardial cavity** – surrounding the heart
 - **Pleural cavities** – surrounding the lungs
 - **Peritoneal cavity** – surrounding the abdominal organs



Sacrococcygeal Teratoma

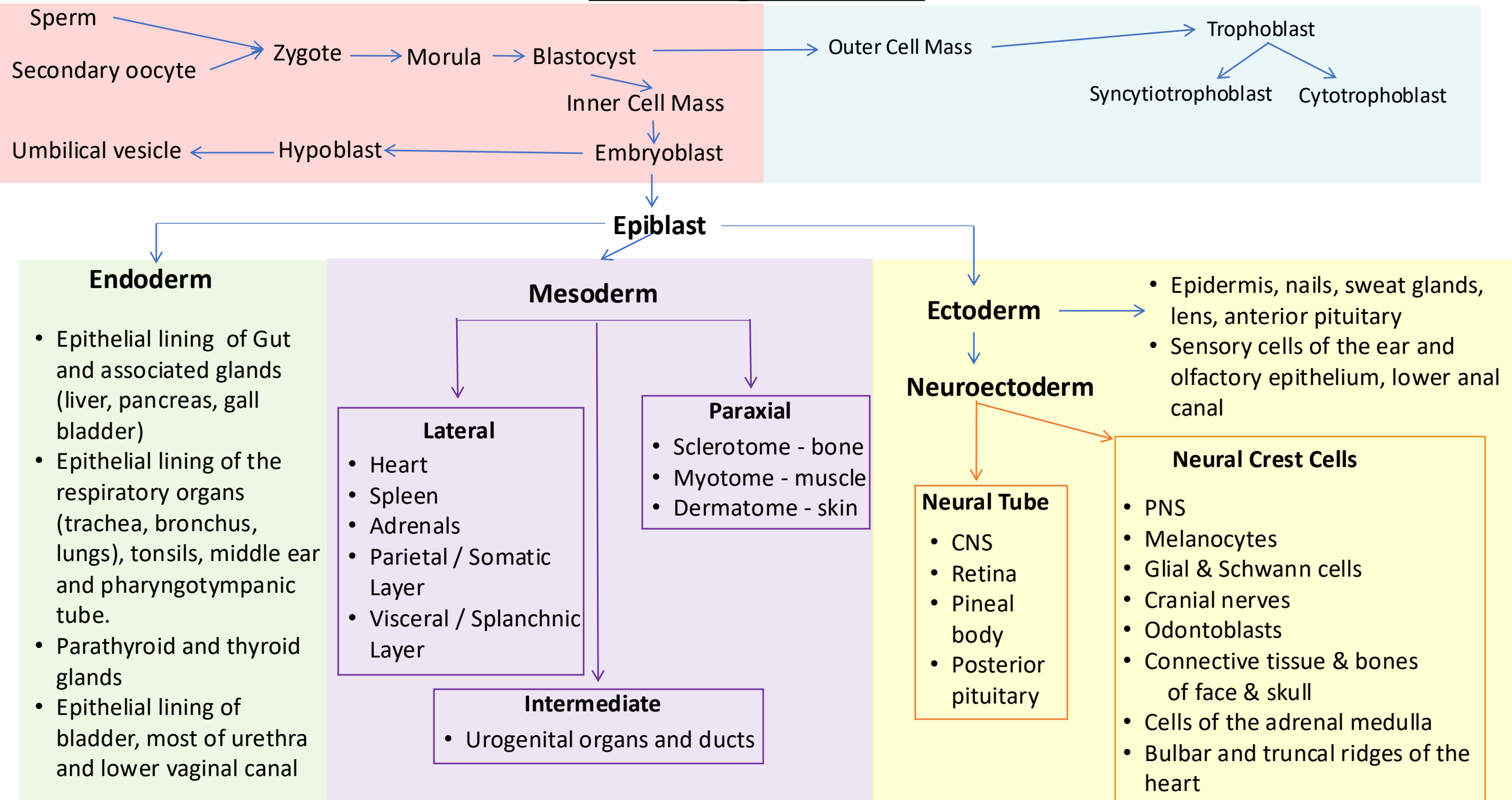
- Most common tumors in newborn infants
- More common in female infants than male
- Persistence of remnants of the pluripotent primitive streak in the caudal region of the embryo
- Derivatives from all three germ layers may be found (teeth, hair, muscle, glands etc.)



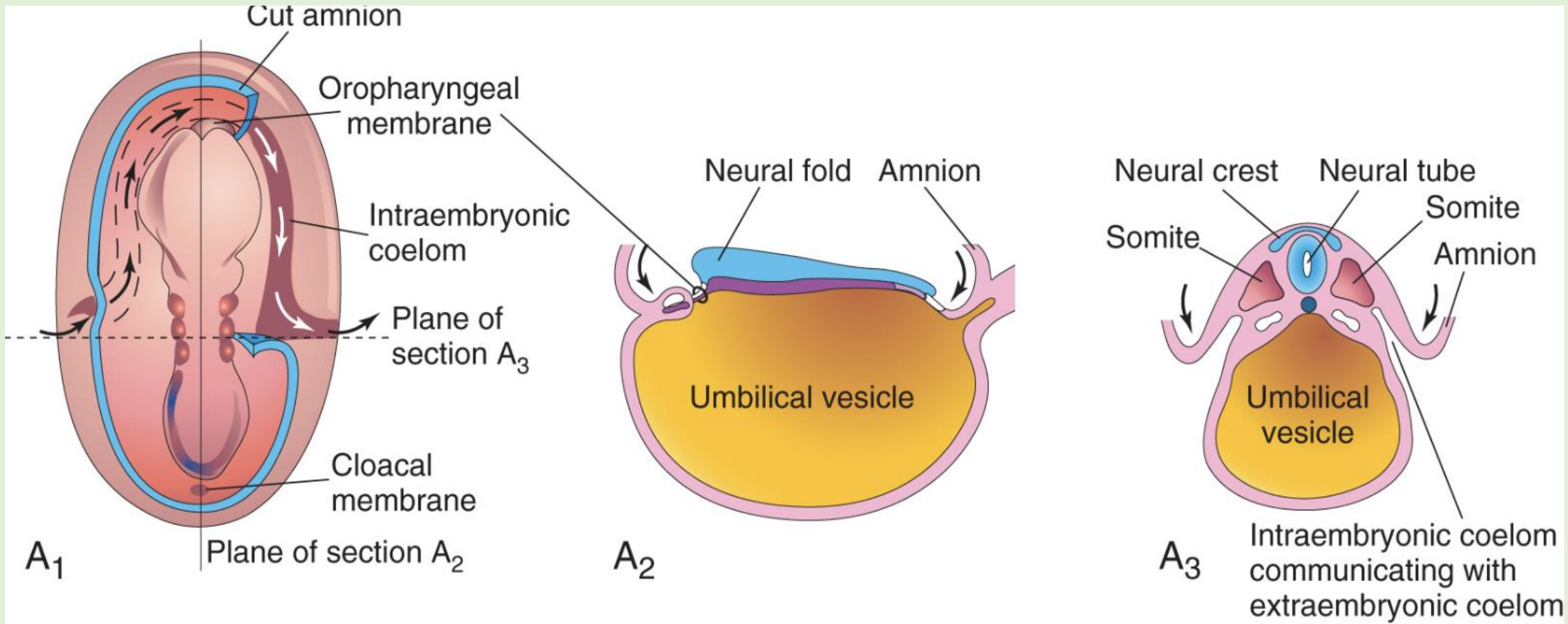
Summary of week 3

- The most significant development is that of the three embryonic cell lines (germ layers) – gastrulation
 - **Ectoderm**
 - **Mesoderm**
 - **Endoderm**
- Many of the processes that start development in week 3 continue on through subsequent weeks
- At this stage if anything harms the developing embryo it will be lost entirely
= **all or nothing period**

The Big picture

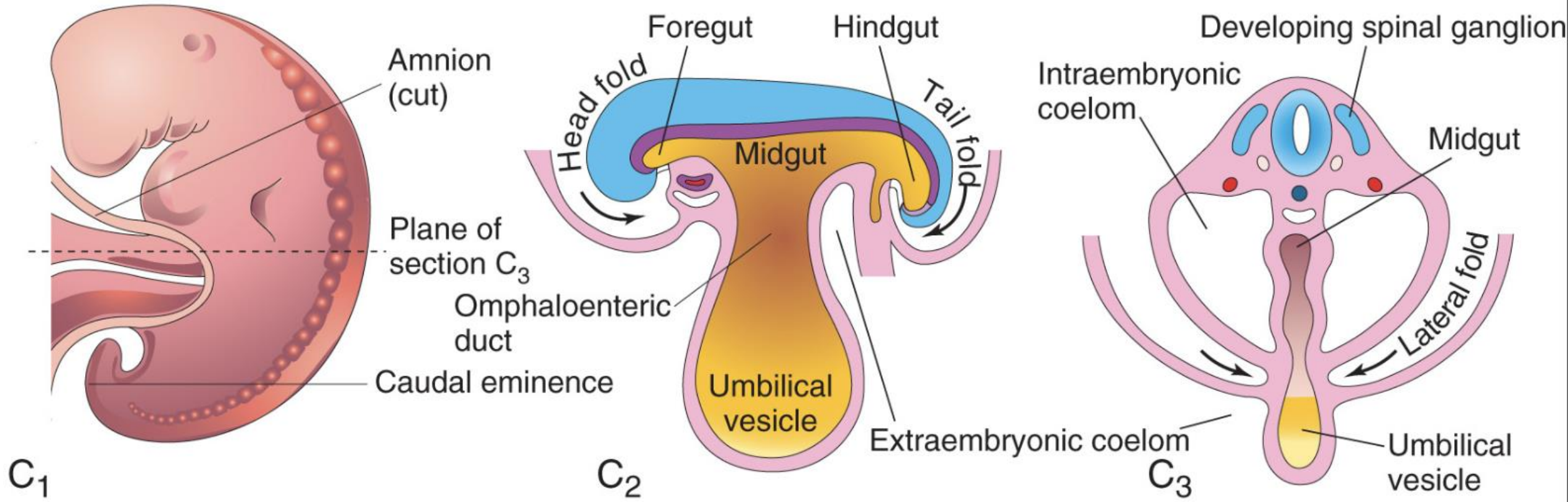


Week 4 - folding



[Video of embryonic folding](#)

Week 4 - folding



[Video of embryonic folding](#)